SMARTCART AI

AI-POWERED CHECKOUT COPILOT

Reducing cart abandonment through behavioral analysis, hesitation detection, and intelligent UX interventions.

REACT · TAILWIND CSS · BROWSER APIS · SIMULATED AI LAYER

Hackathon Submission | May 2026

Frontend Engineering + Product Team

THE CHECKOUT GRAVEYARD

70%

of carts abandoned globally

TOP ABANDONMENT DRIVERS

01 Surprise shipping costs break trust instantly

02 No guidance when users pause or hesitate

03 Coupon hunting — users leave, never return

04 Generic UX ignores individual behavior

> "A user adding products worth **₹4,499** and abandoning at checkout represents real, measurable loss."
>
> At scale, a 10% checkout conversion lift = lakhs in monthly revenue for a growing Indian D2C brand.

70%+	24h	10%
Indian ecommerce on mobile	Window to buy from competitor	Conversion lift = outsized ROI

INTELLIGENCE AT THE EDGE

01 User Interaction Layer

↓

02 Behavioral Signal Collector

↓

03 Hesitation Scoring Engine

↓

04 Intervention Decision Tree

↓

05 React UI + localStorage Persistence

A Frontend-First Checkout Copilot

SmartCart AI layers behavioral intelligence onto standard ecommerce checkout. A lightweight client-side inference engine detects hesitation patterns through DOM event analysis and surfaces targeted interventions — all without external API dependencies.

Zero latency. Zero server costs. Deterministic and debuggable.

INTELLIGENCE AT THE EDGE

TECH STACK

React | Tailwind CSS | Browser APIs | Rules Engine

9 FEATURES ENGINEERED TO CONVERT

AI Checkout Assistant

Contextual help that responds to user behavior via deterministic state machine

Hesitation Detection

Tracks mouse idle, tab switching, scroll patterns, form abandonment

Smart Coupon Engine

Proactively surfaces best applicable coupon — no more hunting

AI Confidence Meter

Visual purchase reassurance aggregating stock, returns, price data

AI Insights Panel

Contextual price trends and social proof at checkout

Smart Recommendations

Cross-sell and upsell based on cart composition

Honest Urgency

Truthful scarcity signals — no fake countdowns

Glassmorphism UI

Translucent layers, subtle blur, depth hierarchy

localStorage Persistence

Cart and coupons survive refreshes, tab switches, browser closes

THE HESITATION DETECTION ENGINE

Technical core — pure JavaScript, zero external ML dependencies

01 CAPTURE — Mouse, scroll, focus, time → **02** DETECT — Tab switch = +25 pts → **03** SCORE — Weighted 0–100 engine → **04** TRIGGER — Threshold breach fires → **05** INTERVENE — Coupon / urgency / help

EXAMPLE INTERVENTION FLOW

Pause 8s on payment, +10 hesitation pts → Switch to another tab, +25 hesitation pts → Score crosses 40, THRESHOLD BREACH → "Save ₹600 with CHECKOUT600", Coupon modal surfaces

WHY THIS APPROACH WINS

Zero latency — runs entirely client-side, no API round-trips

Hackathon-realistic — no training data or GPU needed

Deterministic — trace exactly why every intervention fired

Easily extensible — thresholds in configurable JSON

5 BETS THAT SHAPED THE PRODUCT

BETS WE MADE

01 Frontend-Only Architecture

Invested 80% of time in UX polish and behavioral logic, not backend boilerplate

02 Client-Side Inference Over Cloud ML

Deterministic rules engine — zero API latency, clean migration path later

03 Glassmorphism as Signal

Visual design communicates "this is new" without saying it

04 INR-First Pricing

All demo data in Indian Rupees — grounded in our actual market

05 localStorage Over Backend DB

Persistence without infrastructure — cart survives refreshes and closes

INTENTIONALLY SKIPPED

~~Real payment gateway~~

Simulated payment step — Razorpay/Stripe adds compliance overhead

~~User authentication~~

Anonymous sessions via localStorage — login distracts from checkout

~~Cloud ML backend~~

Local inference demonstrates behavioral logic cleanly

~~Admin dashboard~~

Merchant panel doesn't demo well in 3 minutes

~~Multi-language support~~

English-first for copy quality — Hindi/Tamil in v2

WHAT COMES NEXT

01
Cloud Inference

Rules engine → trained model on aggregated behavior

02
Merchant Dashboard

Abandonment analytics, conversion lift, A/B tests

03
Shopify Plugin

One-click integration for D2C brands

04
Real Payments

Razorpay/Stripe with PCI-compliant tokenization

05
Regional Languages

Hindi, Tamil, Telugu for broader coverage

06
Voice Checkout

Voice input for address and payment on mobile

Each milestone builds on the frontend-first foundation — the rules engine provides a clean interface for swapping in cloud ML later.

CHECKOUT SHOULDN'T BE A GRAVEYARD

Intelligent UX interventions at checkout can measurably improve conversion.

Built by engineers who understand both hackathon constraints and real business pain.

Every feature
addresses a proven abandonment cause

Every decision
prioritizes shipping speed and demo reliability

Zero dependencies
runs entirely in the browser, offline-ready

READY TO DEMO

React · Tailwind CSS · Browser APIs · Simulated AI Layer | May 2026